



Ministry of  
IT & Telecom

**Ignite**  
NATIONAL TECHNOLOGY FUND



**DigiSkills.pk**  
Sara Jahan Hamara

**Hands-on Exercise No. 01**  
**DigiSkills 3.0 Batch-04**  
**Artificial Intelligence using Python**

**Total Marks: 10**

**Due Date: 27/08/2026**

**Instructions:**

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
  - It is submitted after due date.
  - It is not in the required format (.doc or .docx)
  - It does not open, or file is corrupt.
  - It is copied (partial or full) from any source (websites, forums, students, etc.)

**Learning Outcome:**

After completing this exercise, you shall be able to:

- Apply loops and conditional statements to solve simple problems.
- Create and manipulate NumPy arrays.
- Implement basic Object-Oriented Programming (OOP) concepts in Python.

## **Problem Statement**

In this hands-on exercise, you are required to apply your understanding of **Python programming, NumPy, and Object-Oriented Programming (OOP)** to solve practical problems. You will write programs using **loops and conditional statements** to calculate sales tax, use **NumPy** to create and manipulate arrays, and implement a simple **Bank Account System** using OOP concepts.

The objective is to strengthen your programming and problem-solving skills while understanding how different Python concepts work together to build simple real-world applications.

You can use any programming environment such as **Jupyter Notebook, Google Colab, VS Code**, or any other Python editor to write and execute your programs.

After completing each of the following tasks, take screenshots of your **code and its output**, and paste them into a single **MS Word (.doc/.docx)** file as your final solution submission.

### **Tasks:**

1. Suppose a mobile phone shop has 5 sales employees. Write a Python program that uses a for loop to enter the daily sales amount (in Rs.) generated by each employee. Calculate the tax using the following rules: **(Marks 3)**
  - If the sales amount is less than Rs. 50,000, tax is 5%.
  - If the sales amount is between Rs. 50,000 and Rs. 100,000, tax is 10%.
  - If the sales amount is greater than Rs. 100,000, tax is 15%.For each employee, display:
  - Sales Amount
  - Tax AmountAfter entering all 5 employees, display:
  - Total Sales
  - Average Sales
2. Write a Python program using NumPy to: **(Marks 3)**
  - Create a NumPy array containing the marks [78, 85, 92, 67, 88].
  - Display the array.
  - Convert the array into float data type.
  - Display the converted array.
3. Create a Bank Account System using Object-Oriented Programming (OOP). **(Marks 4)**

Create the following classes:

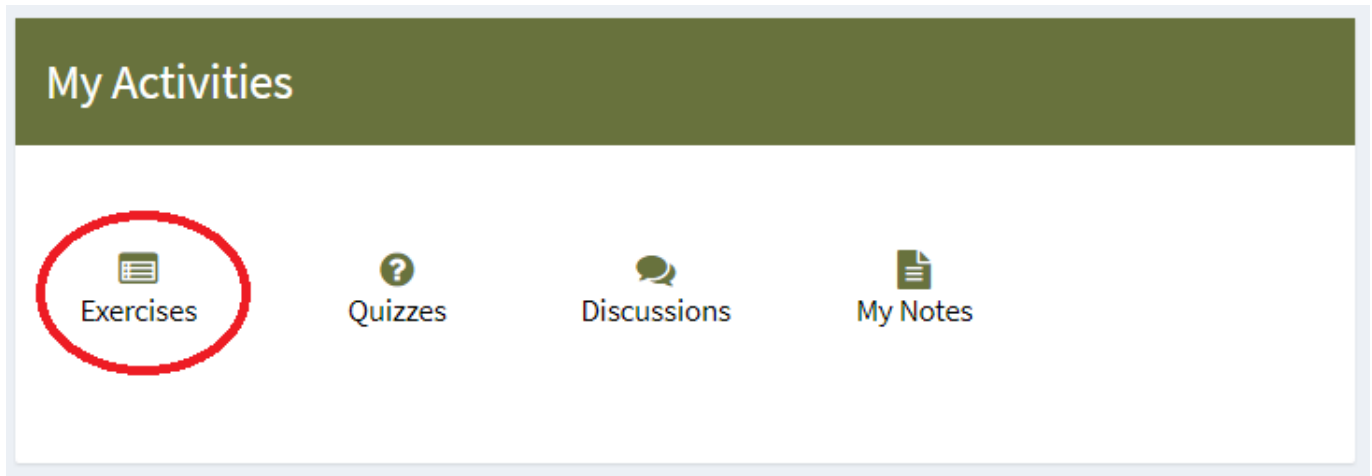
  - Customer
  - AccountThe system should implement the following functionalities:
  - Create a customer account.
  - Deposit money.
  - Withdraw money.
  - Display the current balance.

BEST OF LUCK ☺

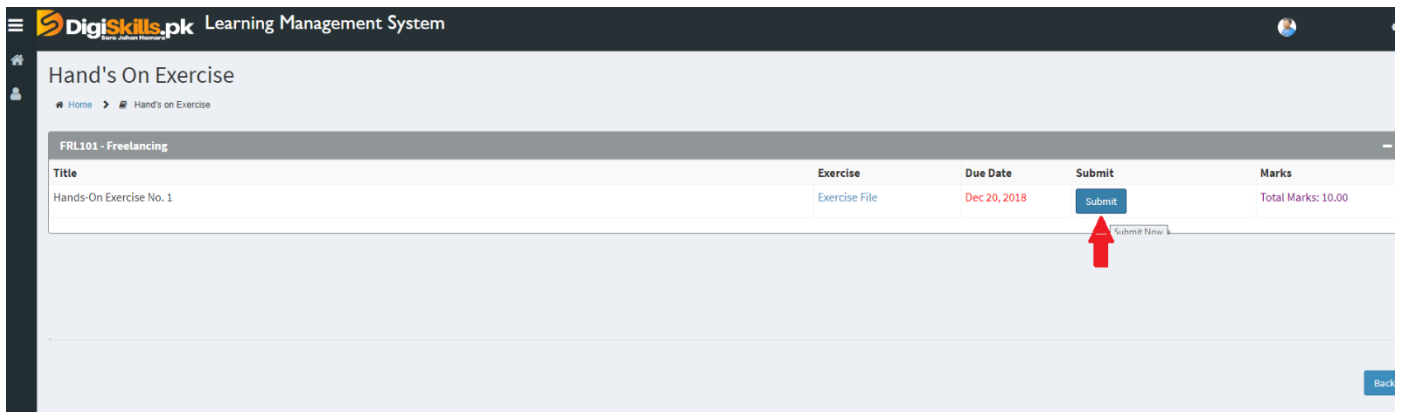
## How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format